

Strainer GAN: Filtering out Impurity Samples in GAN Training

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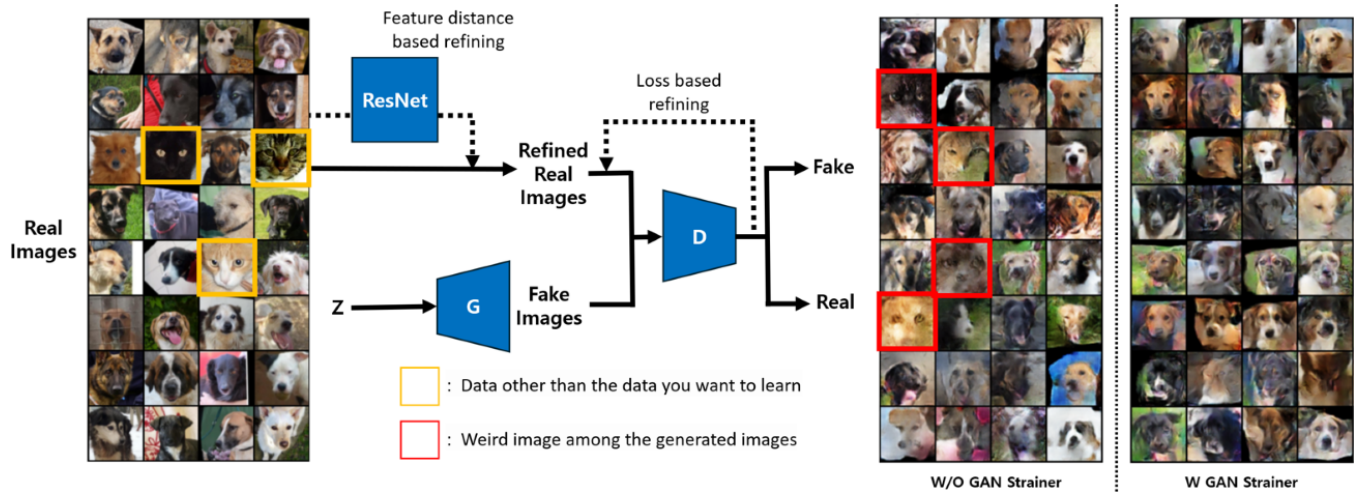


Figure 1: Diagram of Strainer GAN (When trained on impure datasets, DCGAN generates undesired anomalous images, whereas our method successfully produces images consistent with the intended training data)

Abstract

In practical applications of GAN for image generation, large size training samples may be collected from random databases or web-pages. For example, if we collect human face images for training set it may include unexpected impure samples such as cartoon or anime human face images that hinder realistic and high quality human face generation. Manually removing such impure samples is extremely time-consuming and impractical due to the large volume of data. In this work, we propose a novel method to automatically remove impure samples in GAN training.

CCS Concepts

• Computing methodologies → Image generation.

Keywords

Strainer GAN, Impure datasets, Filtering impure samples

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1 Introduction

Generative Adversarial Networks (GANs) have made significant advances in data generation but frequently struggle with noisy real-world datasets. 'Clean' dataset refers to clear, representative samples well aligned with training objectives, while 'impure' dataset includes unexpected or misaligned samples. Manually removing impure samples of large-scale datasets is impractical and costly.

Research on outlier detection and robust GAN training has evolved through various approaches. [Zhang et al. 2021] demonstrated that deep neural networks can learn even randomly labeled data, raising concerns about overfitting to incorrect data. To address the issue, [Jiang et al. 2018] proposed a method to refine dataset identifying noisy label data of higher training loss. [Goldstein and Uchida 2016] evaluated 19 unsupervised outlier detection algorithms across 10 datasets, finding that nearest-neighbor-based methods generally perform well, but most algorithms struggle with high-dimensional data. In GAN-based anomaly detection work, [Akçay et al. 2019] introduced GANomaly, using an encoder-decoder-encoder structure to learn normal data distribution and detect anomalies through reconstruction errors.

Motivated from prior approaches, We propose Strainer GAN automatically minimizing the impact of impure samples. It addresses high-dimensional data challenges by employing feature distance and discriminator loss for outlier and impure sample removal. Our approach is applicable to any kind of GAN architectures

and datasets without requiring pre-training, contributing to the development of more robust and versatile GAN models.

2 Proposed Method

Our approach, Strainer GAN, integrates feature distance-based outlier detection and high-loss impure sample filtering to enhance GAN training on impure datasets. Our method has two steps.

First step is initial feature distance based outlier removal. We begin by refining the initial dataset using a pre-trained ResNet18 to extract features, calculating distances from the mean feature vector, assign Z-scores that a value expresses how many standard deviations a data point is from the mean, and removing outliers exceeding a set threshold. We set the threshold at the midpoint between two key points in the z-score distribution: the point of highest data density and the point where density drops to 0.001%. This filtering ensures the GAN starts learning from a more stable dataset. The training process uses this refined dataset exclusively for the first three epochs, allowing the discriminator to learn primarily from the target data.

Second step is high-loss impure sample filtering. From the fourth epoch onwards, we start each epoch by resetting the dataset refined in the first step. We then further refine this dataset based on the discriminator's loss values. At this stage, we continuously refine the dataset by removing a certain proportion (generally 20%) of real images with the highest losses. This approach prevents initial learning from being distorted by impure samples while enabling finer data refinement as training progresses. To accommodate various contamination scenarios, we allow users to manually adjust both the z-score distribution threshold and the removal ratio.

3 Experiments

This study explores the effectiveness of iterative data refinement techniques in enhancing GAN training robustness on impure datasets. We designate Dog images as the target dataset and use MNIST, CIFAR-10, Cat images as impure datasets to create contaminated Dog datasets. MNIST and CIFAR-10 were used to verify that our method works well with distinctly different datasets, while Cat datasets were employed to test the robustness of our approach in more realistic scenarios where the impure data is similar to the target dataset. The ratio of target dataset to impure dataset was set at 8:2. We trained DCGAN models under two scenarios:

- **DCGAN:** Training on the mixed dataset without refinement
- **Strainer GAN:** Training on a dataset refined using both feature distance and discriminator loss-based methods

Our experiments demonstrate that Strainer GAN significantly outperforms standard DCGAN in reducing anomalous image generation across various scenarios. This improvement was observed when using both distinctly different datasets (MNIST, CIFAR10) and similar face-like images (Cat) as impure data mixed with Dog images. Visual inspection of the generated images supports these findings. Images produced by our proposed models exhibit clearer structures and more consistent styles compared to those from standard DCGAN models. Notably, Strainer GAN rarely generates undesired anomalous images impacted by impure training samples.

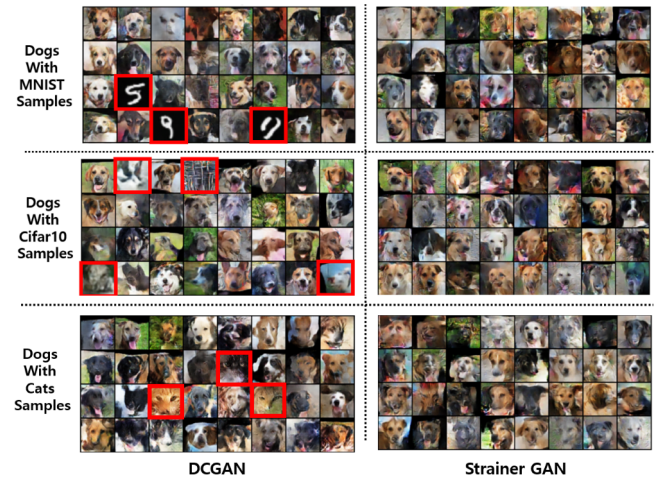


Figure 2: Generated images of GAN trained on impure Dog dataset: For each case, MNIST, CIFAR10 and Cat samples are included in the Dog dataset as impure samples

Our results indicate that the combination of feature distance-based outlier detection and discriminator loss-based refinement is particularly effective in mitigating dataset contamination.

4 Conclusion

This study demonstrates the effectiveness of data refinement in improving the quality of images generated by GAN models trained on impure datasets. The combination of feature distance-based outlier detection and high-loss impure sample filtering proves effective in mitigating the impact of dataset contamination. These findings suggest that such refinement techniques could be valuable in real-world scenarios where dataset purity cannot be guaranteed. As the volume of data continues to grow rapidly, the need for such refinement techniques is likely to increase, and it is expected that this approach will become increasingly relevant in future machine learning applications.

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